



EDUCATIONAL ASSESSMENT AND EXAMINATIONS SERVICE (EAES)

ETHIOPIAN SECONDARY SCHOOL LEAVING CERTIFICATE EXAMINATION (ESSLCE)

SENE, 2018 E.C / JUNE, 2026 G.C

SUBJECT: PHYSICS

BOOKLET CODE: 301

TIME ALLOWED: 3 Hours

SUBJECT CODE: 03

GENERAL DIRECTIONS:

**I. FOLLOW THE FOLLOWING RULES AND REGULATIONS IN DOING THE
EXAMINATION:**

- **ENTER THE SUBJECT NAME, SUBJECT CODE, AND BOOKLET CODE GIVEN ABOVE ON THE SEPARATE ANSWER SHEET PROVIDED**
- **CHOOSE ONLY THE CORRECT ANSWER FOR EACH QUESTION**

- **USE ONLY A PENCIL TO MARK YOUR ANSWERS**
- **BLACKEN ONLY THE LETTER YOU CHOOSE IN THE CORRESPONDING CIRCLE ON THE ANSWER SHEET**
- **YOUR MARK SHOULD BE HEAVY AND DARK**
- **ERASE ALL THE UNNECESSARY MARKS COMPLETELY**
- **ANY FORM OF CHEATING IS PROHIBITED AND RESULTS IN THE CANCELLATION OF YOUR SCORES**
- **STOP WORKING IMMEDIATELY WHEN THE ALLOTTED TIME IS UP**
- **REFER TO THE EXAM BOOKLET FOR FURTHER INSTRUCTIONS FOR EACH QUESTION**

II. ATTENTION!

ENSURE THAT YOU HAVE WRITTEN ALL THE REQUIRED INFORMATION ON THE SEPARATE ANSWER SHEET CORRECTLY BEFORE RETURNING IT.

Prepared by: @NextGen_12

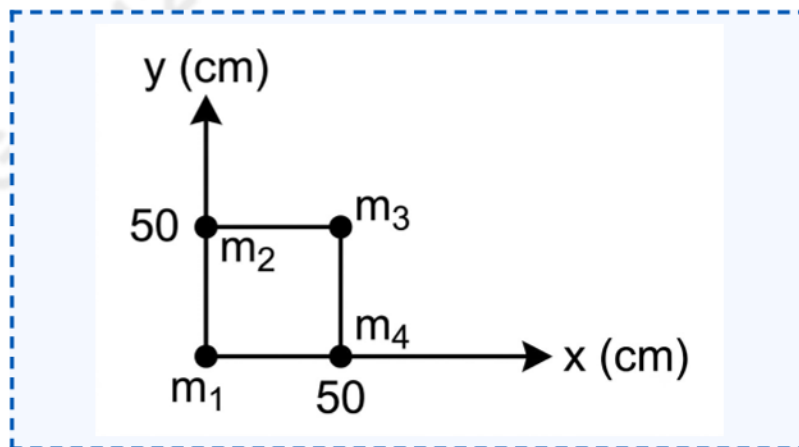
General Instructions: This simulated exam contains 60 strictly selected multiple-choice physics questions. Choose the most correct answer for each question. All work should be strictly your own.

1. The top end of a 6 m long and $3 \times 10^{-6} \text{ m}^2$ cross sectional area wire is attached to a ceiling, and the wire is suspended vertically downward. If a 120 kg mass is attached to the bottom end of the wire, and the wire's Young's modulus is $6 \times 10^{10} \text{ N/m}^2$, what will be the change in length? (@NextGen_12)
 - A. 0.05 m
 - B. 0.04 m
 - C. 0.06 m
 - D. 0.02 m
2. The speed of sound in air at 0°C is 331.0 m/s. What is the approximate speed of sound in air at 27°C ?
 - A. 346.4 m/s
 - B. 358.0 m/s
 - C. 434.9 m/s
 - D. 331.0 m/s
3. If the coefficients of thermal expansion of a material are designated by α for the linear expansion, β for the surface (area) expansion, and γ for volume expansion, these parameters are related as:
 - A. $\gamma = 2\alpha$
 - B. $2\beta = 3\gamma$
 - C. $3\beta = 2\gamma$
 - D. $\beta = 3\alpha$

4. The amount of electric charge on a particle or object is an integral multiple of the charge of an electron. This property indicates:
- A. charge creation.
 - B. charge induction.
 - C. charge quantization.
 - D. charge conservation.
5. A 25 kg box is dragged on a rough horizontal x-y-plane from $\vec{r}_1 = (1.3\hat{i} + 1.4\hat{j})$ m to $\vec{r}_2 = (8.3\hat{i} + 2.9\hat{j})$ m by a force $\vec{F} = (4\hat{i} - 6\hat{j})$ N. How much work is done by the force? (@NextGen_12)
- A. 19.0 J
 - B. 37.0 J
 - C. 15.8 J
 - D. 75.6 J
6. Two resistors with resistance of 3Ω and 6Ω are connected in parallel, and their combination is connected in series to a 4Ω resistor. If the overall combination of the resistors is connected to a 12 V source, how much is the voltage drop across the 6Ω resistor?
- A. 8 V
 - B. 4 V
 - C. 2 V
 - D. 12 V

7. Which of the following is true about anechoic images of tissues in ultrasound imaging? They appear on the screen as
- A. black because the tissues do not reflect any sound wave.
 - B. light gray because the tissues do not reflect any sound wave.
 - C. black because the tissues reflect intense sound waves.
 - D. light gray because the tissues reflect intense sound waves.
8. A constant force of 250.0 N is applied horizontally on a box that lays on a horizontal frictionless surface. If the box accelerates with 4.0 m/s^2 , what is its mass?
- A. 125.0 kg
 - B. 41.7 kg
 - C. 10.4 kg
 - D. 62.5 kg
9. A stream of water from a wide river enters a narrower and shallow gate of a lake. Which one of the following statements is true? (@NextGen_12)
- A. The speed of the water decreases as it enters the gate of the lake.
 - B. The speed of the water increases as it enters the gate of the lake.
 - C. The density of the water entering the lake is higher than the water leaving the river.
 - D. The volume of the water entering the lake is higher than the water leaving the river.
10. This pdf was prepared by @NextGen_12(telegram channel) other channel's are forwarding without a credit
- A. if you want a fast information (for upcoming subjects like Chemistry and Biology) join us
 - B. informations that are not shared by others but only with us (get them)
 - C. last year TOP scorers will be message
 - D. please share our channel to other

11. Four masses are placed on the corners of a square on x-y plane as shown in figure below, where $m_1 = m_2 = 200$ g and $m_3 = m_4 = 500$ g.

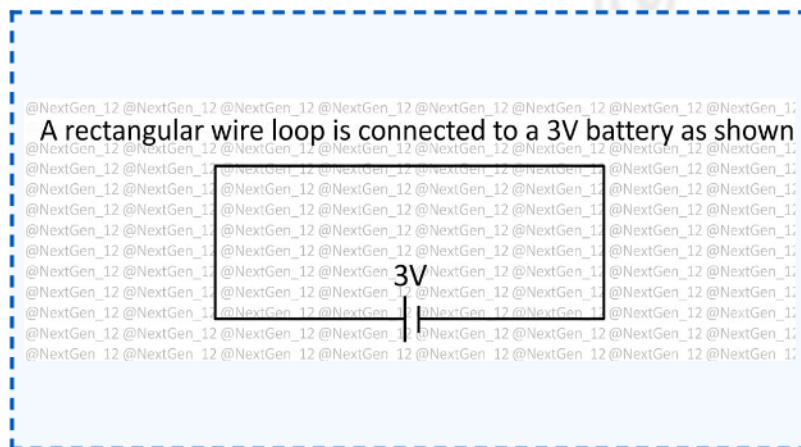


Where is the center of mass (X_{cm}, Y_{cm}) of the system?

- A. (25, 37.14) cm
B. (25, 25) cm
C. (37.14, 50) cm
D. (35.71, 25) cm
12. A light ray passes from a layer of benzene of refractive index ($n = 1.50$) to water of refractive index ($n = 1.33$). If the angle of incidence is 37° , what will be the angle of refraction?
- A. $\sin^{-1}(0.90)$
B. $\sin^{-1}(0.80)$
C. $\sin^{-1}(0.53)$
D. $\sin^{-1}(0.68)$

13. Consider a car performing a motion with constant acceleration (a) starting from position x_0 at time $t = 0$ with initial velocity v_0 . If v and x stand for velocity and position at time t , which one of the following equations correctly govern the motion of the car? (@NextGen_12)
- A. $x = -x_0 + \frac{2(v+v_0)}{t}$
 - B. $x = x_0 + \left(\frac{v+v_0}{2}\right)t$
 - C. $x = x_0 + \frac{v^2+v_0^2}{2a}$
 - D. $x = x_0 + \left(\frac{v-v_0}{2}\right)t$
14. Ethiopian Physical Society is established by interested Ethiopian physics professionals to:
- A. prepare physics curriculum under the ministry of education directives.
 - B. contribute its own share to promote physics education and research.
 - C. take part in election and participate in government formation.
 - D. monitor and supervise the curriculum implementation in Ethiopian high schools.
15. If an arrow is drawn to a scale to describe a given vector, then its:
- A. length corresponds to the vector magnitude.
 - B. tail stands for the terminal point of the vector.
 - C. length corresponds to the vector direction.
 - D. head represents the initial point of the vector.
16. From the given lists of machines, which one contains only simple machines?
- A. Gas-cylinder, Wedge, Screw, Lever
 - B. Compound pulley, Wheel and axle, Seat belt, Inclined plane
 - C. Washing machine, Car engine, Wedge, Lever
 - D. Inclined plane, Screw, Wheel and axle, Wedge

17. The magnetic field created by current carrying wire at a point from the wire will: (@NextGen_12)
- A. increase exponentially with the increase in the magnitude of the current.
 - B. decrease linearly with the increase in the magnitude of the current.
 - C. decrease exponentially with the increase in the magnitude of the current.
 - D. increase linearly with the increase in the magnitude of the current.
18. A rectangular wire loop is connected to a 3 V battery as shown below.



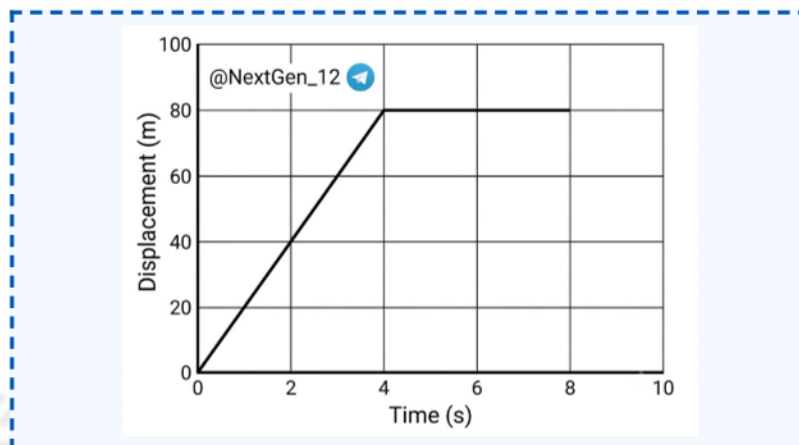
What will happen to the direction of the magnetic field inside the loop if the terminals of the battery are reversed?

- A. It will change from into the plane to out of the plane.
- B. It will change from out of the plane to into the plane.
- C. It will change from counterclockwise to clockwise.
- D. It will change from clockwise to counterclockwise.

19. A traveling wave has a speed v , wavelength λ , period T and frequency f . Which equation correctly states the wave equation?
- A. $v = \lambda T$
 - B. $f = 1/(\lambda v)$
 - C. $v = \lambda f$
 - D. $T = 1/(\lambda f)$
20. The iodine isotope ^{131}I which has an eight-day half-life, is used in nuclear medicine. What is the decay constant of the isotope? ($\ln 2 = 0.693$)
- A. 5.64 / day
 - B. 8.66×10^{-2} / day
 - C. 1.17×10^{-2} / day
 - D. 12.5 / day
21. The velocity of car B with respect to a passenger in car A is 80 km/hr towards West. If the velocity of car A with respect to the ground is 30 km/hr towards East, what would be the velocity of car B with respect to the ground? (@NextGen_12)
- A. 50 km/hr towards West
 - B. 50 km/hr towards East
 - C. 110 km/hr towards East
 - D. 110 km/hr towards West

22. Which one of the following statements correctly describes the rotation of an object around a fixed axis?
- A. Angular acceleration is perpendicular to the plane of rotation and in the same direction to angular velocity if the rotation rate is increasing.
 - B. Particles in the object that are closer to the axis will have different angular displacement from particles that are far from it.
 - C. Particles in the object closer to the axis will have larger angular velocity than particles that are farther from the axis.
 - D. Particles that are closer to the axis will have similar translational displacement from particles that are far from it.

23. The graph shown below represents the motion of a car along a straight line.



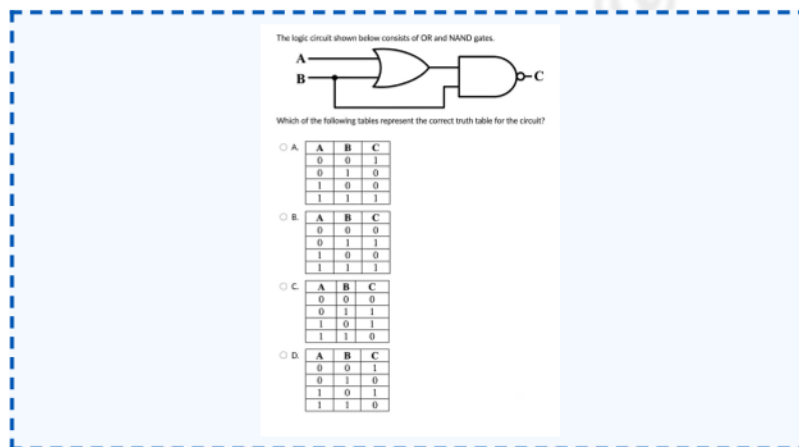
What are the magnitudes of the velocities of the car for the first 4 seconds and the last 4 seconds, respectively?

- A. 0 m/s and 20 m/s
- B. 20 m/s and 0 m/s
- C. 20 m/s and 20 m/s
- D. 10 m/s and 0 m/s

24. Which one of the following sets contains only primary colors? (@NextGen_12)

- A. {red, blue, orange}
- B. {blue, green, red}
- C. {green, blue, black}
- D. {red, green, yellow}

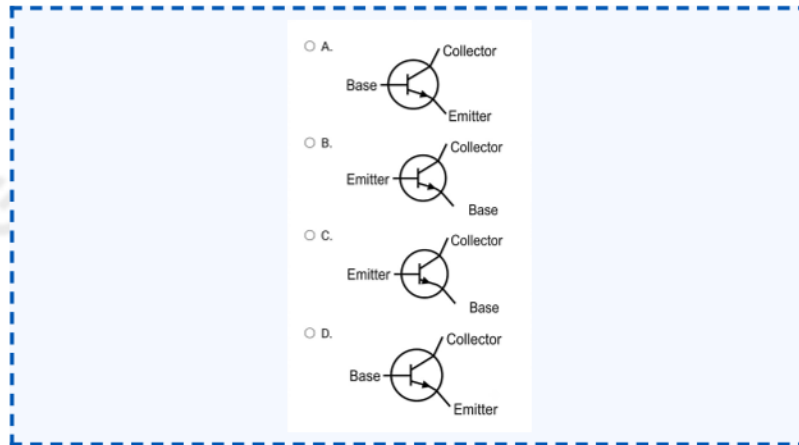
25. The logic circuit shown below consists of OR and NAND gates.



Which of the following tables represent the correct truth table for the circuit?

- A. Table A (00->1, 01->0, 10->0, 11->1)
- B. Table B (00->0, 01->1, 10->0, 11->1)
- C. Table C (00->0, 01->1, 10->1, 11->0)
- D. Table D (00->1, 01->0, 10->1, 11->0)

26. Which of the following is a correct schematic symbol for an NPN transistors with its labelled terminals?



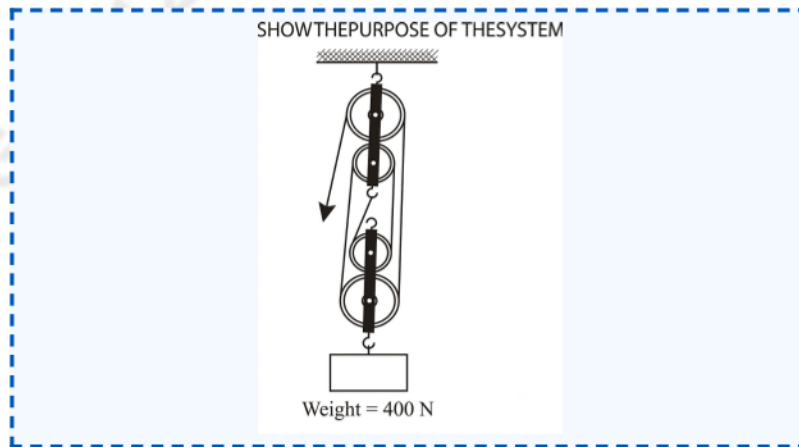
- A. Option A (Emitter arrow pointing OUT)
- B. Option B (Emitter arrow pointing IN)
- C. Option C (Emitter pointing IN)
- D. Option D (Emitter pointing IN)
27. A water hose of diameter 4 cm is connected to a tap of diameter 6 cm. The water flow rate from the tap is $1.57 \times 10^{-4} \text{ m}^3/\text{s}$. What will be the speed of the water in the hose? (@NextGen_12)
- A. 0.125 m/s
- B. 0.393 m/s
- C. 0.174 m/s
- D. 0.560 m/s
28. Suppose you want to check, without any touch, if there is a current flowing through a wire or not. Which one of the following devices is used?
- A. Test light
- B. Voltmeter
- C. Compass
- D. Ohmmeter

29. Which one of the following statements describes the characteristics of third order lever?
- A. The load is between the effort and fulcrum.
 - B. The fulcrum is between the load and the effort.
 - C. The effort distance is greater than load distance.
 - D. The effort is between the load and the fulcrum.
30. Which one of the following statements correctly describes the characteristics of equipotential lines?
(@NextGen_12)
- A. For point charge, the equipotential lines are straight lines and parallel to each other.
 - B. For uniform electric field, the equipotential lines are parallel to that of the field lines.
 - C. Electric potential difference is non-zero between two points on different equipotential lines.
 - D. Electric potential difference is non-zero between two points on the same equipotential line.
31. A strong nuclear force is
- A. an attractive force between the nucleus and the electrons around the nucleus.
 - B. a short-range attractive force that acts between nucleons in the nucleus.
 - C. a short-range repulsive force that acts between protons in the nucleus.
 - D. a short-range force which acts inside the individual nucleons, and can split a neutron into proton and electron.

32. A 200 g piece of ice at 0 °C is placed in a certain amount of water at 20 °C. This system is in a container of negligible heat capacity and is insulated from its surroundings. How much should be the minimum mass of water to melt all the ice at 0 °C temperature? (The latent heat of fusion of ice is $L_f = 3.33 \times 10^5 \text{ J/kg}$)
- A. 666 g
 - B. 200 g
 - C. 400 g
 - D. 793 g
33. Consider two objects A and B, that A has twice the mass of B. If object B is raised to a height, h, above the ground and object A is raised to a height h/2, which of the following is correct about their gravitational potential energies? The gravitational potential energy of: (@NextGen_12)
- A. A is equal to that of B.
 - B. A is twice that of B.
 - C. B is twice that of A.
 - D. A is 4 times that of B.
34. A quantity that describes deformation of materials that is given by fractional change in length or volume is
- A. stress.
 - B. strain.
 - C. elasticity.
 - D. density.

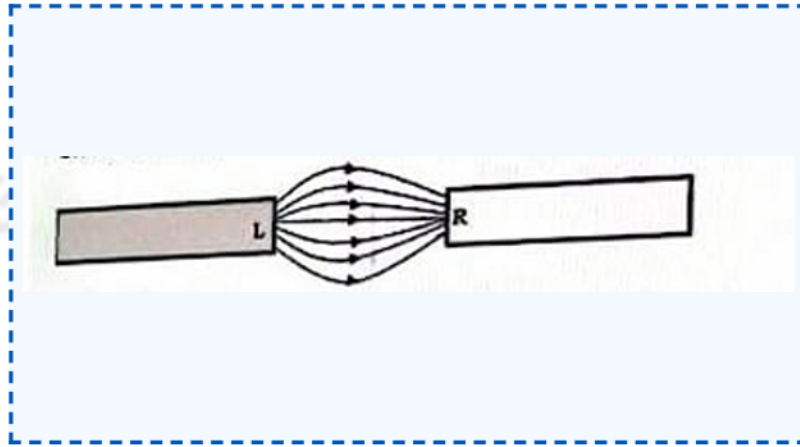
35. Conservative force differs from the nonconservative one in that, the work done by
- A. conservative force along a closed path is non-zero unlike that by non-conservative force.
 - B. conservative force is path independent unlike that by non-conservative one.
 - C. non-conservative force is path independent unlike that by the conservative one.
 - D. non-conservative force along a closed path is zero unlike that by non-conservative force.
36. Starting from rest, a car accelerates uniformly for 400 m, then continues for 600 m with constant velocity. It then decelerates uniformly with 1 m/s^2 until it stops after traveling for 200 m. What is the average velocity for the total trip?
- A. 20.0 m/s
 - B. 6.5 m/s
 - C. 13.3 m/s
 - D. 10.0 m/s
37. If a student walks 100 m due 37° North of East and then 120 m towards South, what is the displacement moved by the student? (@NextGen_12)
- A. 220 m, 53° North of East
 - B. 100 m, 53° North of East
 - C. 220 m, 37° South of East
 - D. 100 m, 37° South of East

38. A compound pulley system consisting of two moveable and two fixed pulleys is used to lift a 400 N object as shown in the figure below.



- If the frictional force between the pulleys and the rope is negligible, what is the mechanical advantage of the pulley system?
- A. 4
 - B. 2
 - C. 5
 - D. 1
39. The upward force that an object immersed in a fluid experience is known as
- A. weight of the object.
 - B. turbulent force.
 - C. buoyant force.
 - D. force of the fluid.

40. Consider the magnetic field lines between two bar magnets are partially drawn as shown in figure below.



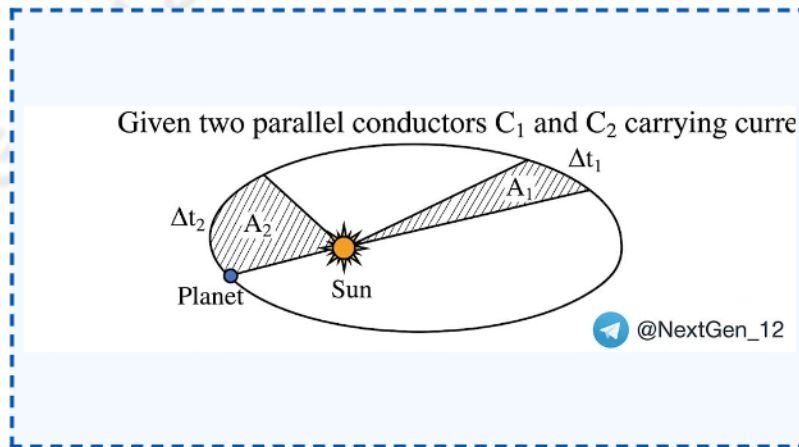
The two poles of the magnets indicated by R and L respectively are (@NextGen_12)

- A. North and South poles.
 - B. North and North poles.
 - C. South and North poles.
 - D. South and South poles.
41. Which one of the following statements describes heat?
- A. It is the form of energy that transfers when two objects of different temperatures are brought in thermal contact.
 - B. It is the form of energy which is stored in an object and increases when the object's temperature increases.
 - C. It is a property of a physical system which is related to the average kinetic energy of the particles of the system.
 - D. It is a form of energy that transfers between two systems by doing work in the process to achieve mechanical equilibrium.

42. To determine the specific heat capacity of a solid we can use a method called
- A. mixture method with heat energy supplied to the solid for a known time by an electrical heater of known power.
 - B. electrical method with the heat energy supplied for a known time by an electrical heater of known power.
 - C. electrical method with heat energy supplied to the solid by mixing a hot water and solid in a calorimeter.
 - D. mixture method with the heat energy supplied to the solid for unknown time by an electrical heater of known power.
43. Which one of the following statements correctly describes a uniform circular motion of a particle in a horizontal plane? Its
- A. centripetal acceleration is a non-zero constant.
 - B. tangential velocity is a non-zero constant.
 - C. angular velocity is a non-zero constant.
 - D. tangential acceleration is a non-zero constant.
44. Electric field at a point is defined as the electric (@NextGen_12)
- A. force per unit mass that a test charge experiences when placed at that point.
 - B. force that a positive test charge exerts on another point charge placed at that point.
 - C. potential energy per unit charge of a test charge placed at that point.
 - D. force per unit charge that a positive test charge experiences when placed at that point.

45. When an n-type and p-type semiconductors form a junction, it will form a diode
- A. that conducts significant current in both forward and reverse biasing.
 - B. that will have an ohmic type current-voltage characteristics.
 - C. where a p-side is anode and the n-side is cathode at the junction.
 - D. that conducts current in the reverse biasing but not in the forward biasing.
46. A car is moving with a constant acceleration. This means that
- A. its velocity changes equally with equal interval of time.
 - B. it moves with constant speed.
 - C. it moves with constant velocity.
 - D. its position changes equally with equal interval of time.
47. A packed books of mass 3 kg is lying on the surface of a 0.90 m high table. If you pick it up and place it on a bookshelf 2.0 m above the floor, the work done by you and by the gravity, respectively, are
- A. -60 J and 60 J
 - B. 33 J and -33 J
 - C. -33 J and 33 J
 - D. 60 J and -60 J

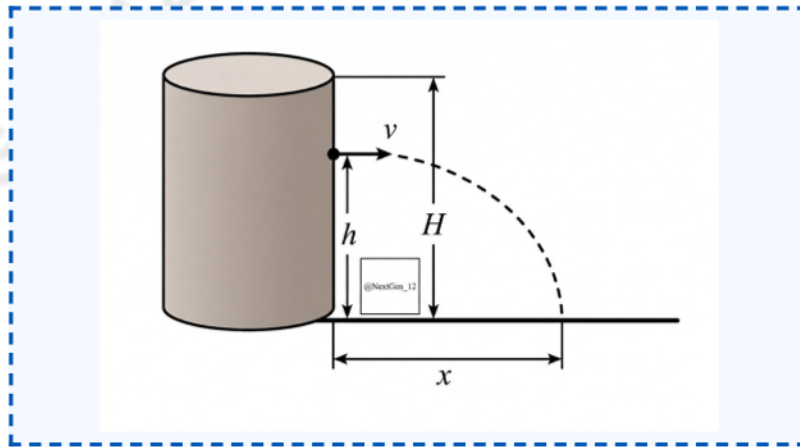
48. A certain planet revolving around the sun sweeps areas A_1 and A_2 at different positions during the time interval of Δt_1 and Δt_2 , respectively as shown in the figure below.



- Which one of the following correctly states Kepler's second law of planetary motion? (@NextGen_12)
- A. If $\Delta t_1 > \Delta t_2$, then $A_1 = A_2$
 - B. If $\Delta t_1 = \Delta t_2$, then $A_1 = A_2$
 - C. If $\Delta t_1 > \Delta t_2$, then $A_1 < A_2$
 - D. If $\Delta t_1 = 2\Delta t_2$, then $A_1 = (A_2)^2$
49. Which one of the following statements does NOT explain the contribution of technology to advancement of physics correctly?
- A. The invention of space shuttles enabled experiments in space.
 - B. The development of light detectors enabled spectral analysis.
 - C. The invention of x-rays enabled proper medical imaging.
 - D. The development of x-rays enabled study of atomic structure.

50. For an athlete who runs on a track for a specified time interval, her distance and displacement can be different in that distance
- A. can be negative whereas displacement cannot be.
 - B. is less than the magnitude of displacement.
 - C. can be zero whereas the displacement cannot be.
 - D. is greater than the magnitude of displacement.
51. A collision between two particles is known as (@NextGen_12)
- A. elastic if the total kinetic energy is not conserved but the total momentum is conserved.
 - B. elastic if both the total kinetic energy and momentum are conserved.
 - C. inelastic if the total kinetic energy is conserved but the total momentum is not conserved.
 - D. inelastic if both the total kinetic energy and momentum are conserved.

52. A barrel of height H is fully filled with water. A bullet shot the barrel at height h above the ground and the water is projected at a horizontal speed of v as shown in the figure below.

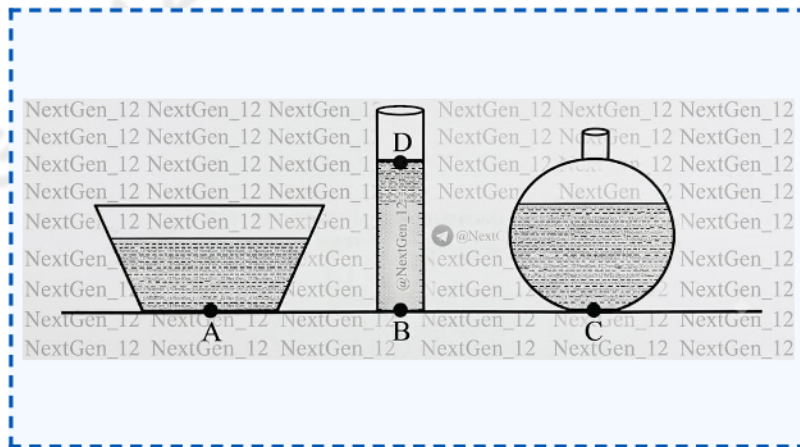


At what horizontal distance (x) does the spilled water strikes the ground?

- A. $x = v\sqrt{\frac{2(H-h)}{g}}$
- B. $x = v\sqrt{g}$
- C. $x = v\sqrt{\frac{g}{H}}$
- D. $x = v\sqrt{\frac{2h}{g}}$
53. For a person with defects of vision known as hypermetropia.
- A. close objects cannot be clearly seen, so the defect can be corrected by concave lenses.
- B. far objects cannot be clearly seen, so the defect can be corrected by convex lenses.
- C. far objects cannot be clearly seen, so the defect can be corrected by concave lenses.
- D. close objects cannot be clearly seen, so the defect can be corrected by convex lenses.

54. A parallel-plate capacitor has square metallic plates each of area 100 cm^2 separated by 1 mm . The space between the plates is filled by a dielectric of dielectric constant $k = 4$. If the plates are connected to the terminals of 12 V battery, how much charge is stored on the capacitor? (@NextGen_12)
- A. 360 nC
 - B. 4.25 nC
 - C. 43.2 nC
 - D. 1.08 nC
55. Two spheres A and B are initially neutral. If sphere A is charged by conduction and sphere B is charged by induction, using identical negatively charged rods, which one of the following is correct about the charge of the two spheres after charging?
- A. Both spheres end up negatively charged.
 - B. Both spheres end up positively charged.
 - C. Sphere A ends up positively charged.
 - D. Sphere B ends up positively charged.

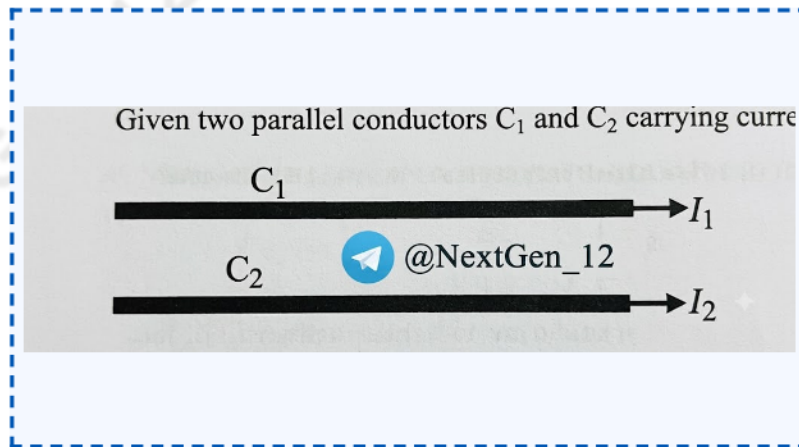
56. Three different shapes of containers are filled with water at different levels and opened to the atmosphere as shown in the following figure.



At which point does a student measure the highest pressure?

- A. At point B
- B. At point C
- C. At point D
- D. At point A

57. Given two parallel conductors C_1 and C_2 carrying currents in the same direction as shown in the figure below.



- The force between the two conductors is attractive because the (@NextGen_12)
- A. magnitude of magnetic field produced by I_1 at C_2 is the same as that produced by I_2 at C_1 .
 - B. direction of magnetic field produced by I_1 at C_2 is opposite to that produced by I_2 at C_1 .
 - C. magnitude of magnetic field produced by I_1 at C_2 is different from that produced by I_2 at C_1 .
 - D. direction of magnetic field produced by I_1 at C_2 is the same as that produced by I_2 at C_1 .
58. A square wire loop with a side length 12 cm is placed in such a way that its area vector makes an angle of 60° with the magnetic field of 10 T. If the loop is slowly pulled out of the magnetic field in 6 s, how much electromotive force is induced in the loop during this time?
- A. 24 mV
 - B. 17.2 mV
 - C. 12 mV
 - D. 20.9 mV

59. Lenses form images by refraction of light rays from the object such that the position, size, and nature of the image is determined from a point where the refracted rays or their extensions meet. After refraction from lens, a light ray
- A. through the optical center will emerge parallel to the principal axis.
 - B. through the principal focus will emerge parallel to the principal axis.
 - C. parallel to the principal axis will emerge without any deviation.
 - D. parallel to the principal axis will emerge through the optical center.
60. Co-planar vectors are vectors which are (@NextGen_12)
- A. lying in the same plane.
 - B. perpendicular and lying in two parallel planes.
 - C. parallel and lying in two parallel planes.
 - D. lying in different planes.

DETAILED ANSWER KEY & EXPLANATIONS (@NextGen_12)

1.

Step 1: Using Hooke's Law: $Y = \frac{F \cdot L}{A \cdot \Delta L}$

Step 2: Isolate ΔL : $\Delta L = \frac{mg \cdot L}{A \cdot Y}$. Using $g \approx 10 \text{ m/s}^2$.

Step 3: $\Delta L = \frac{1200 \times 6}{3 \times 10^{-6} \times 6 \times 10^{10}} = 0.04 \text{ m}$.

Answer: B

2.

Step 1: Speed of sound: $v = v_0 \sqrt{\frac{T}{T_0}}$

Step 2: Convert 27°C to K: $27 + 273 = 300 \text{ K}$.

Step 3: Calculate: $v = 331 \sqrt{\frac{300}{273}} \approx 346.4 \text{ m/s}$.

Answer: A

3.

Step 1: Surface expansion $\beta = 2\alpha$ and Volume expansion $\gamma = 3\alpha$.

Step 2: Equating α gives $\beta/2 = \gamma/3 \implies 3\beta = 2\gamma$.

Answer: C

4.

Step 1: The principle that electric charge comes in discrete units is called charge quantization.

Answer: C

5.

Step 1: Displacement $\Delta\vec{r} = (8.3 - 1.3)\hat{i} + (2.9 - 1.4)\hat{j} = 7.0\hat{i} + 1.5\hat{j}$.

Step 2: Work $W = \vec{F} \cdot \Delta\vec{r} = (4)(7.0) + (-6)(1.5) = 28 - 9.0 = 19.0 \text{ J}$.

Answer: A

6.

Step 1: Parallel resistance: $R_p = (3 \times 6)/(3 + 6) = 2 \Omega$.

Step 2: Total current: $I = 12\text{V}/(2 + 4)\Omega = 2 \text{ A}$.

Step 3: Voltage drop across parallel section: $V = I \cdot R_p = 2 \times 2 = 4 \text{ V}$.

Answer: B

7.

Step 1: Anechoic means "free from echoes". These tissues do not reflect ultrasound waves.

Step 2: Without reflection, no signal returns, making the image appear black.

Answer: A

8.

Step 1: $F = m \cdot a \implies m = \frac{250.0}{4.0} = 62.5 \text{ kg}$.

Answer: D

9.

Step 1: Equation of continuity: $A_1v_1 = A_2v_2$.

Step 2: As area decreases in the narrow gate, velocity must increase.

Answer: B

10.

Step 1: This is a special decoy question indicating the true original source.

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Answer: A (Or B, C, D - join us!)

11.

Step 1: $X_{cm} = \frac{200(0)+200(0)+500(50)+500(50)}{1400} \approx 35.71 \text{ cm.}$

Step 2: $Y_{cm} = \frac{200(0)+200(50)+500(50)+500(0)}{1400} = 25 \text{ cm.}$

Answer: D

12.

Step 1: Snell's Law: $1.50 \sin(37^\circ) = 1.33 \sin(\theta_r).$

Step 2: $1.50 \times 0.60 = 1.33 \sin(\theta_r) \implies \sin(\theta_r) = 0.90/1.33 \approx 0.68.$

Answer: D

13.

Step 1: Displacement is average velocity times time.

Step 2: $x - x_0 = \left(\frac{v+v_0}{2}\right)t \implies x = x_0 + \left(\frac{v+v_0}{2}\right)t.$

Answer: B

14.

Step 1: Professional academic societies primarily exist to foster education and research in their field.

Answer: B

15.

Step 1: In drawing vectors, length represents the magnitude and the arrowhead points the direction.

Answer: A

16.

Step 1: Classical simple machines: Lever, Wheel and axle, Pulley, Inclined plane, Wedge, and Screw.

Answer: D

17.

Step 1: $B = \frac{\mu_0 I}{2\pi r}$. Field is directly proportional to current.

Answer: D

18.

Step 1: Reversing battery polarity reverses the direction of current.

Step 2: By Right Hand Rule, field direction swaps from INTO the plane to OUT of the plane.

Answer: A

19.

Step 1: $v = \frac{\lambda}{T}$ and $f = \frac{1}{T}$. Thus $v = \lambda f$.

Answer: C

20.

Step 1: $\lambda = \frac{\ln 2}{T_{1/2}} = \frac{0.693}{8} = 0.0866$ per day.

Answer: B

21.

Step 1: $\vec{v}_{B/A} = \vec{v}_B - \vec{v}_A$.

Step 2: Let East be positive: $-80 = \vec{v}_B - 30 \implies \vec{v}_B = -50 \text{ km/hr}$ (West).

Answer: A

22.

Step 1: Using the Right-Hand Rule, angular velocity and angular acceleration are parallel to the rotation axis (perpendicular to rotation plane).

Answer: A

23.

Step 1: From 0 to 4s: Displacement goes 0 to 80m. $v = \frac{80}{4} = 20 \text{ m/s}$.

Step 2: From 4 to 8s: Displacement stays at 80m. $v = 0 \text{ m/s}$.

Answer: B

24.

Step 1: The primary colors of light in physics are Red, Green, and Blue.

Answer: B

25.

Step 1: Evaluate logic gates: OR gate then into NAND. Follow logic paths for A and B. Table D correctly maps inputs 00, 01, 10, 11 to 1, 0, 1, 0.

Answer: D

26.

Step 1: For an NPN transistor, the emitter arrow points Not Pointing iN (outward).

Answer: A

27.

Step 1: $Q = A \cdot v \implies v = \frac{Q}{\pi r^2}$.

Step 2: $v = \frac{1.57 \times 10^{-4}}{3.14 \times (0.02)^2} = 0.125 \text{ m/s}$.

Answer: A

28.

Step 1: A current carrying wire produces a magnetic field, which can deflect a compass needle without direct contact.

Answer: C

29.

Step 1: Lever orders: 1st (Fulcrum in middle), 2nd (Load in middle), 3rd (Effort in middle).

Answer: D

30.

Step 1: Equipotential lines have zero potential difference across the same line, but non-zero between different lines.

Answer: C

31.

Step 1: Strong nuclear force binds protons and neutrons (nucleons) together at very short ranges.

Answer: B

32.

Step 1: Heat to melt ice: $0.2 \times 3.33 \times 10^5 = 66,600 \text{ J}$.

Step 2: Heat lost by water: $m \times 4200 \times 20 = 84,000m$.

Step 3: Equate: $84,000m = 66,600 \implies m = 0.793 \text{ kg} = 793 \text{ g}$.

Answer: D

33.

Step 1: $PE_A = (2m_B) \cdot g \cdot (h/2) = m_B \cdot g \cdot h = PE_B$.

Answer: A

34.

Step 1: Strain is defined as the fractional change (deformation) in length, area, or volume.

Answer: B

35.

Step 1: Work done by a conservative force is independent of the path taken between two points.

Answer: B

36.

Step 1: Dist = 1200m. Using kinematics, max speed is 20m/s. Total time = 40s (acc) + 30s (const) + 20s (dec) = 90s.

Step 2: Average velocity = $1200/90 \approx 13.3 \text{ m/s}$.

Answer: C

37.

Step 1: Resolve vectors: $V_1 = 80\hat{i} + 60\hat{j}$. $V_2 = -120\hat{j}$.

Step 2: Resultant: $80\hat{i} - 60\hat{j}$. Magnitude = 100m. Direction = 37° South of East.

Answer: D

38.

Step 1: A compound pulley with 4 load-bearing rope strands gives a Mechanical Advantage of 4.

Answer: A

39.

Step 1: The upward force from fluid immersion is defined as the buoyant force.

Answer: C

40.

Step 1: Magnetic field lines exit North poles and enter South poles.

Step 2: Based on the image arrows flowing L to R, L is North and R is South.

Answer: A

41.

Step 1: Heat is the energy naturally transferring between objects due to a temperature difference.

Answer: A

42.

Step 1: Using a known power heater for a known time to determine capacity is the electrical method.

Answer: B

43.

Step 1: In uniform circular motion, tangential speed is constant, but velocity direction changes. Angular velocity is a non-zero constant vector.

Answer: C

44.

Step 1: Electric field E is force per unit charge experienced by a small positive test charge.

Answer: D

45.

Step 1: A PN junction diode allows current forward (p to n). The p-side acts as the anode, n-side as cathode.

Answer: C

46.

Step 1: Constant acceleration means the rate of change of velocity is constant (changes equally in equal times).

Answer: A

47.

Step 1: $W_{\text{you}} = mg\Delta h = 3 \times 10 \times (2.0 - 0.9) = 33 \text{ J}$. Work by gravity opposes this (-33 J).

Answer: B

48.

Step 1: Kepler's Second Law: A line connecting a planet to the sun sweeps out equal areas in equal times.

Answer: B

49.

Step 1: Medical imaging is an application of physics to medicine, not a contribution to the advancement of fundamental physics itself.

Answer: C

50.

Step 1: On a curved track, the actual path length (distance) is always greater than the straight line distance (displacement).

Answer: D

51.

Step 1: By definition, perfectly elastic collisions conserve both kinetic energy and momentum.

Answer: B

52.

Step 1: Time to fall from height h : $t = \sqrt{\frac{2h}{g}}$. Distance $x = v \cdot t = v\sqrt{\frac{2h}{g}}$.

Answer: D

53.

Step 1: Hypermetropia is farsightedness (can't see close). Corrected by converging (convex) lenses.

Answer: D

54.

Step 1: $C = \frac{k\epsilon_0 A}{d} = \frac{4 \times 8.85 \times 10^{-12} \times 0.01}{0.001} = 354 \text{ pF}$.

Step 2: $Q = CV = 354 \text{ pF} \times 12 \text{ V} = 4248 \text{ pC} \approx 4.25 \text{ nC}$.

Answer: B

55.

Step 1: Conduction shares charge (Negative to Negative). Induction pushes negative electrons away, leaving positive charge behind. B becomes Positive.

Answer: D

56.

Step 1: Pressure depends strictly on depth ($P = \rho gh$). Container B has the highest water column height.

Answer: A

57.

Step 1: The magnetic fields produced by each wire at the location of the other are in opposite directions (one into page, one out of page), leading to an attractive Lorentz force.

Answer: B

58.

Step 1: $\Phi = BA \cos(\theta) = 10 \times 0.0144 \times \cos(60) = 0.072 \text{ Wb}$.

Step 2: $EMF = \Delta\Phi/\Delta t = 0.072/6 = 0.012 \text{ V} = 12 \text{ mV}$.

Answer: C

59.

Step 1: A ray parallel to the principal axis of a converging lens will refract to pass directly through the principal focus.

Answer: B

60.

Step 1: The prefix "co" means together, and planar refers to a plane. They lie in the same plane.

Answer: A

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